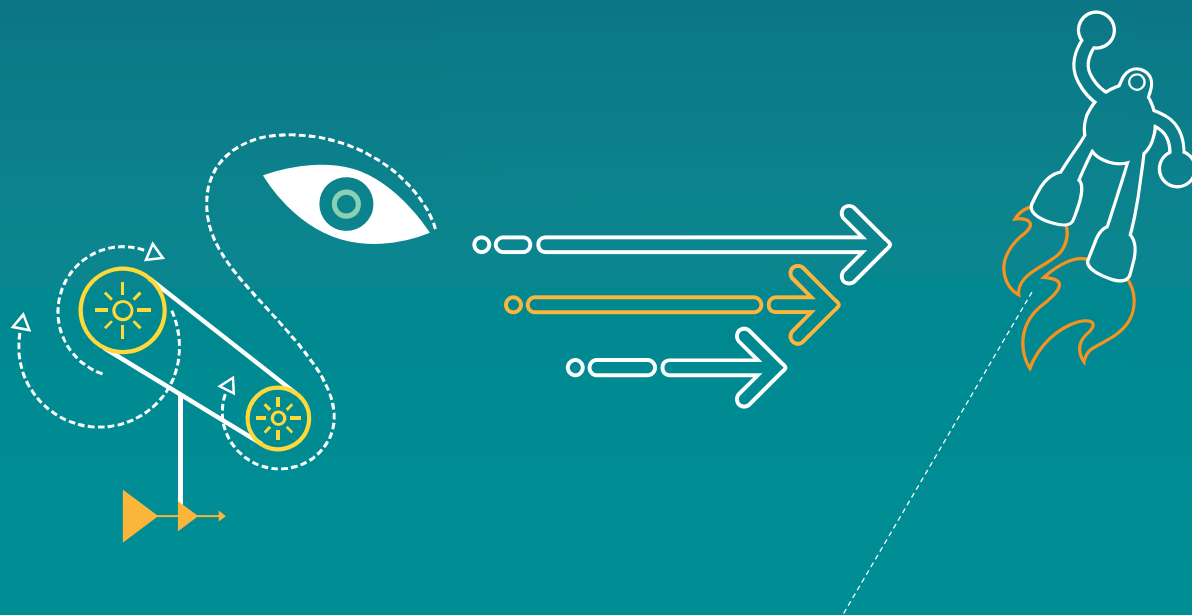
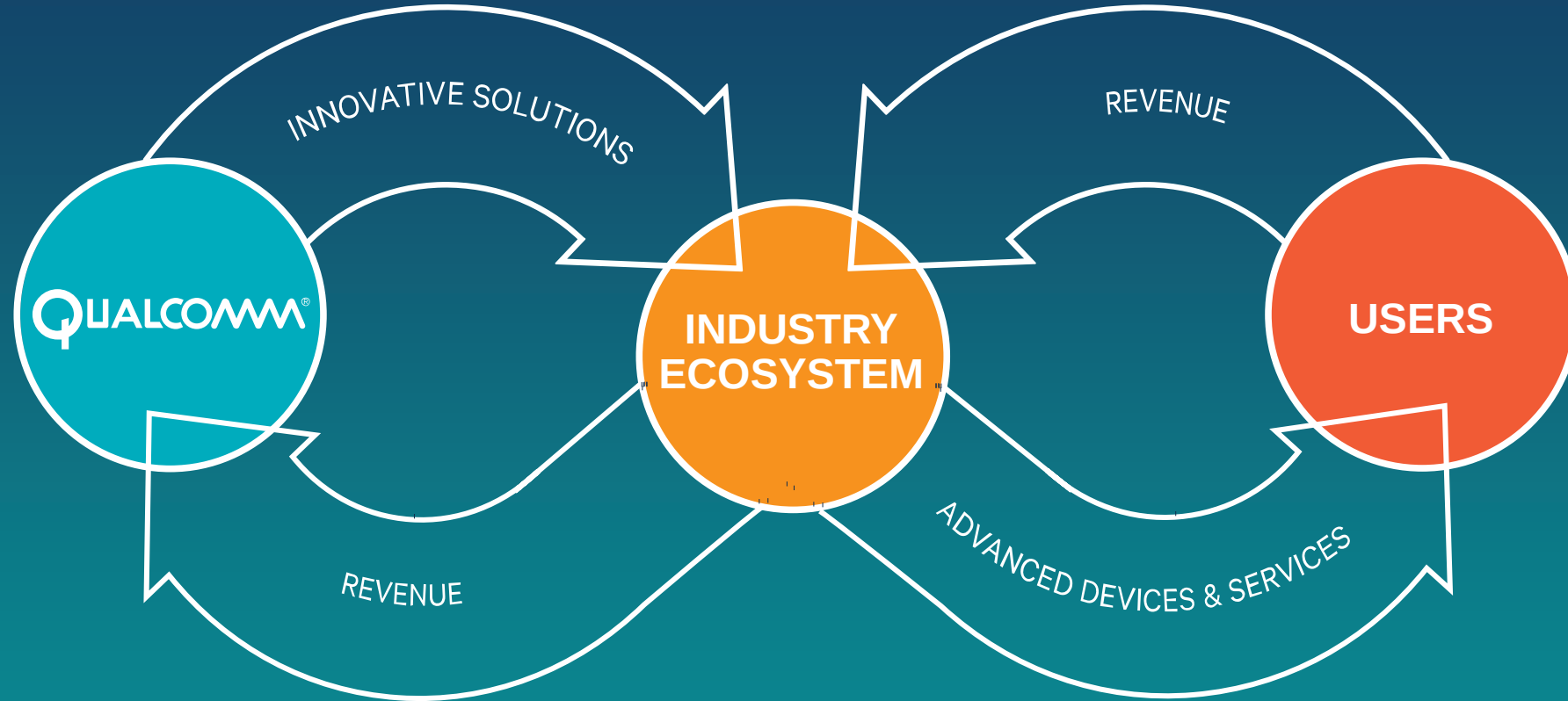


Stein Lundby
November 18, 2015

Creating Value-Added Products Using Surface Electronics



A Model for Innovation



**Complex
Systems R&D
Leadership**



**Design for
Manufacturability**

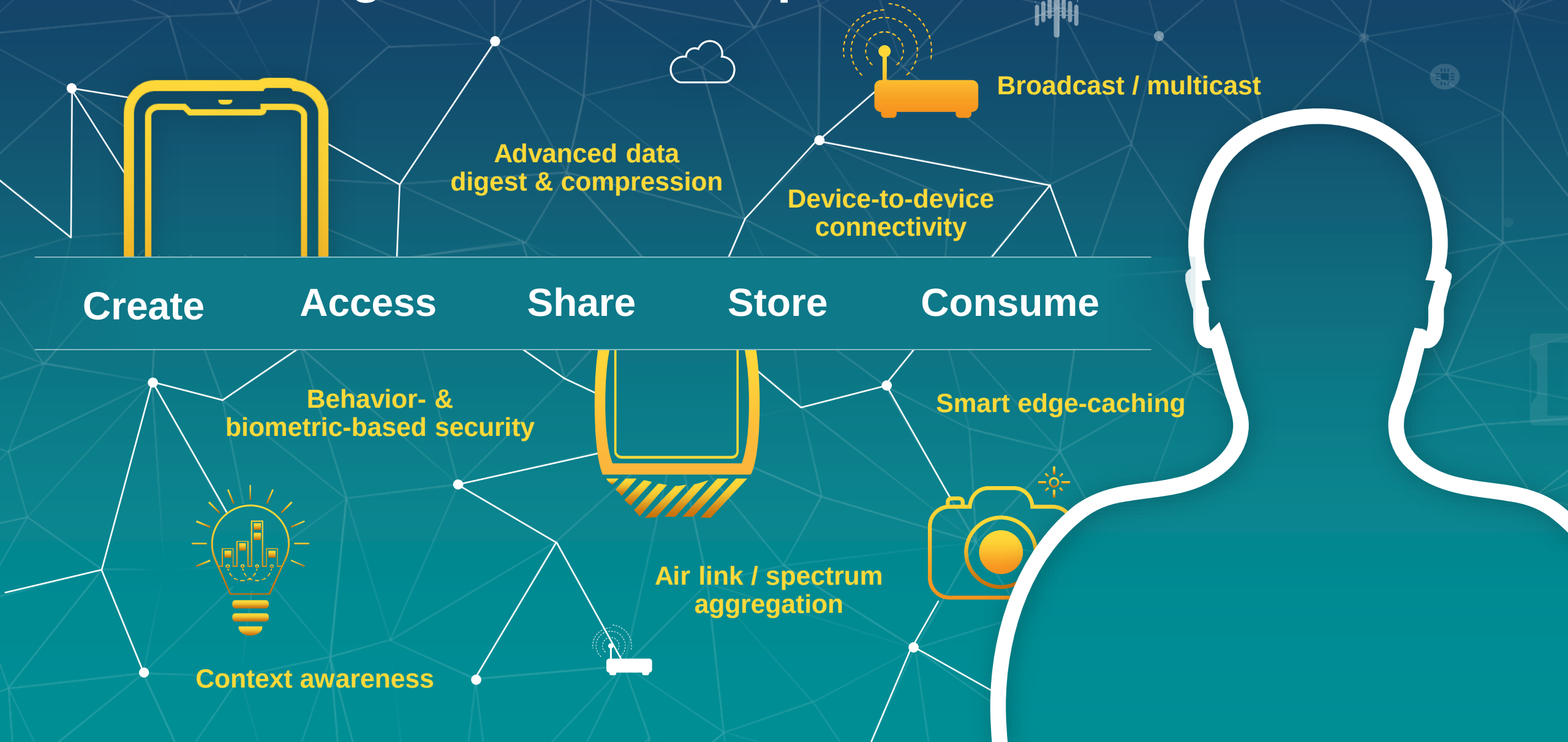


**Supply Chain
Management**



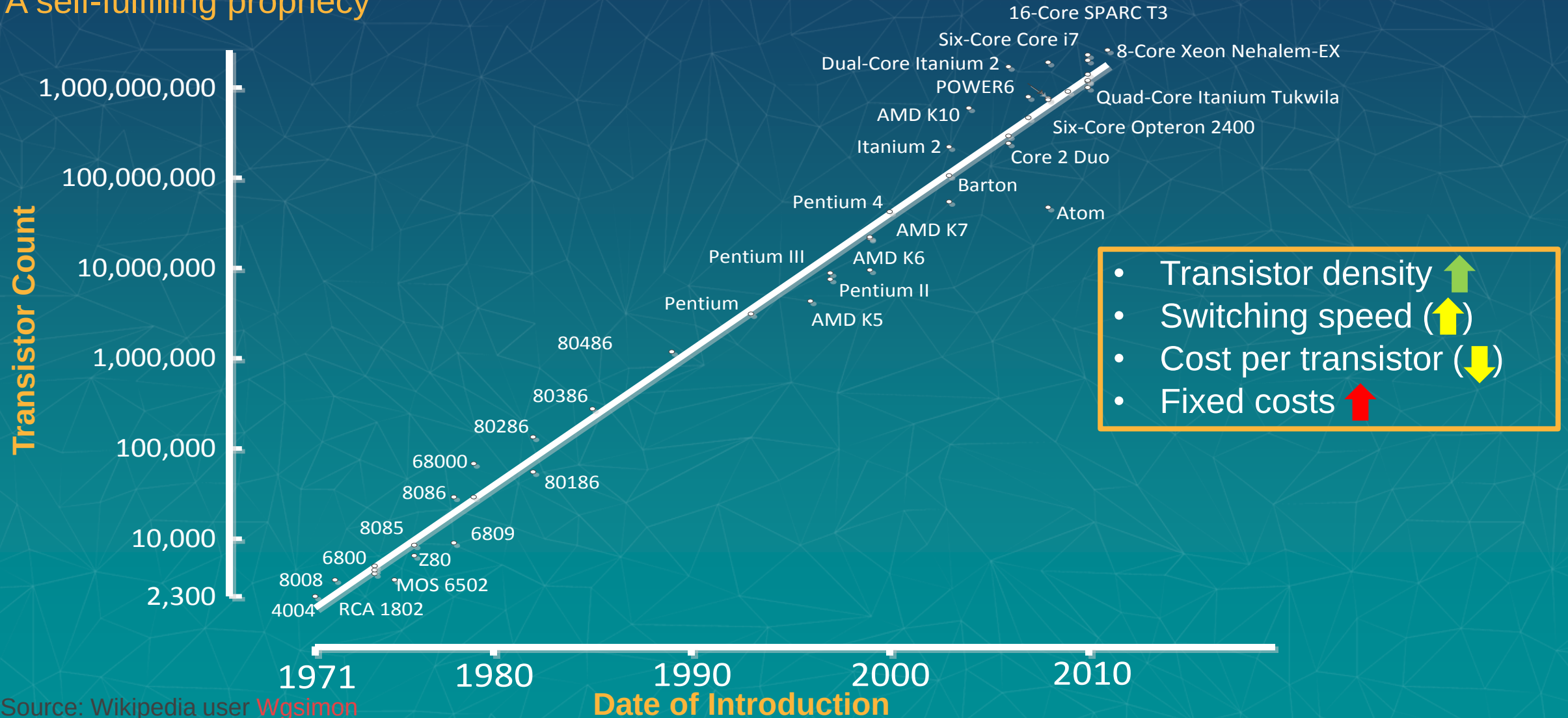
**Customer Support
Engineering**

Transforming our Relationship with Data



Chasing Moore's Law

A self-fulfilling prophecy

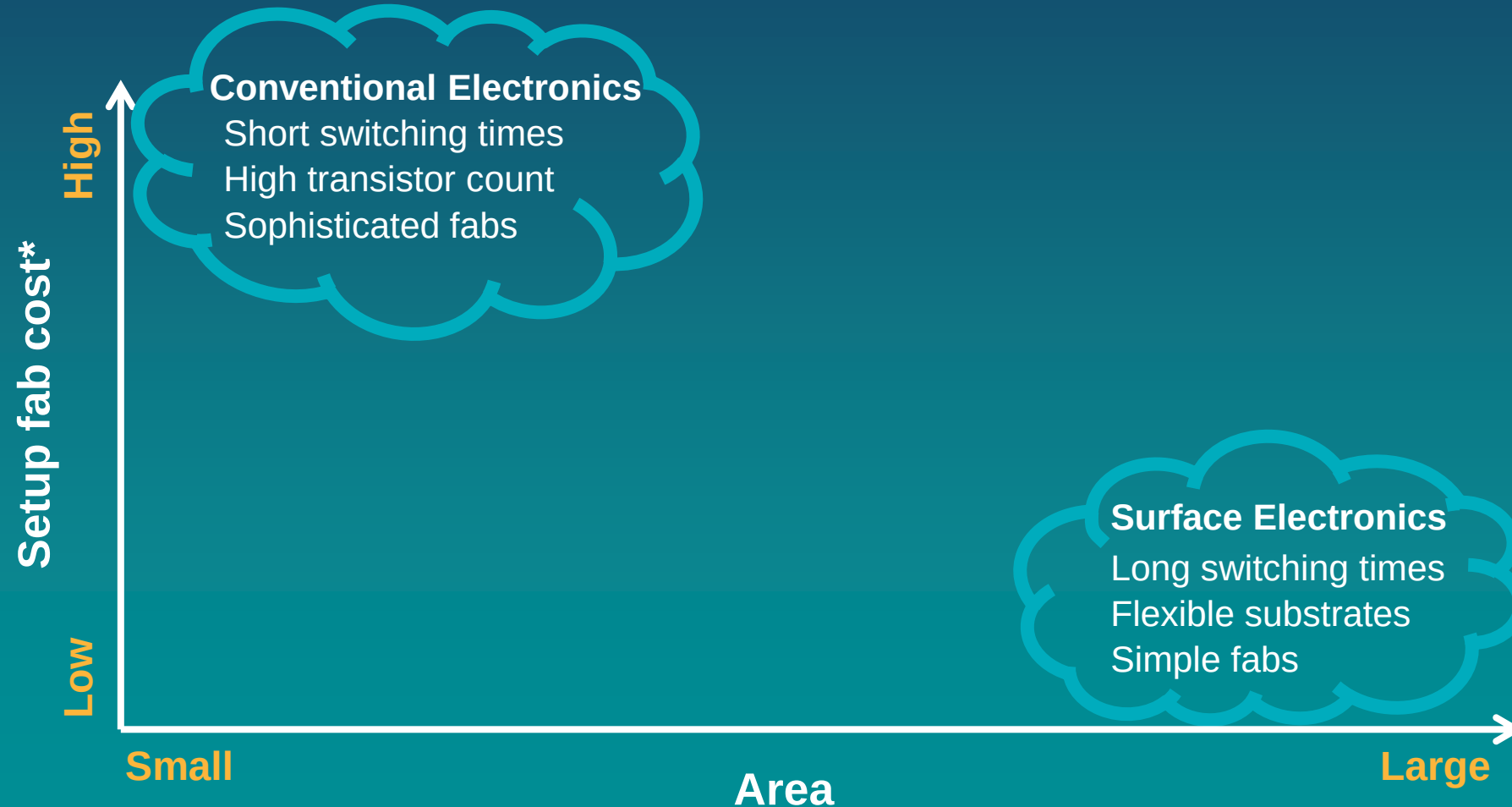


Source: Wikipedia user [Wgsgimon](#)

Surface Electronics (Printed, Deposited, Flexible, etc.)

A different approach to electronics

Cost vs Area





Perspective on Surface Electronics

- **Opportunity**

- Surface electronics offer a paradigm shift in electronics
- Significant potential in dimensions not addressed by traditional electronics

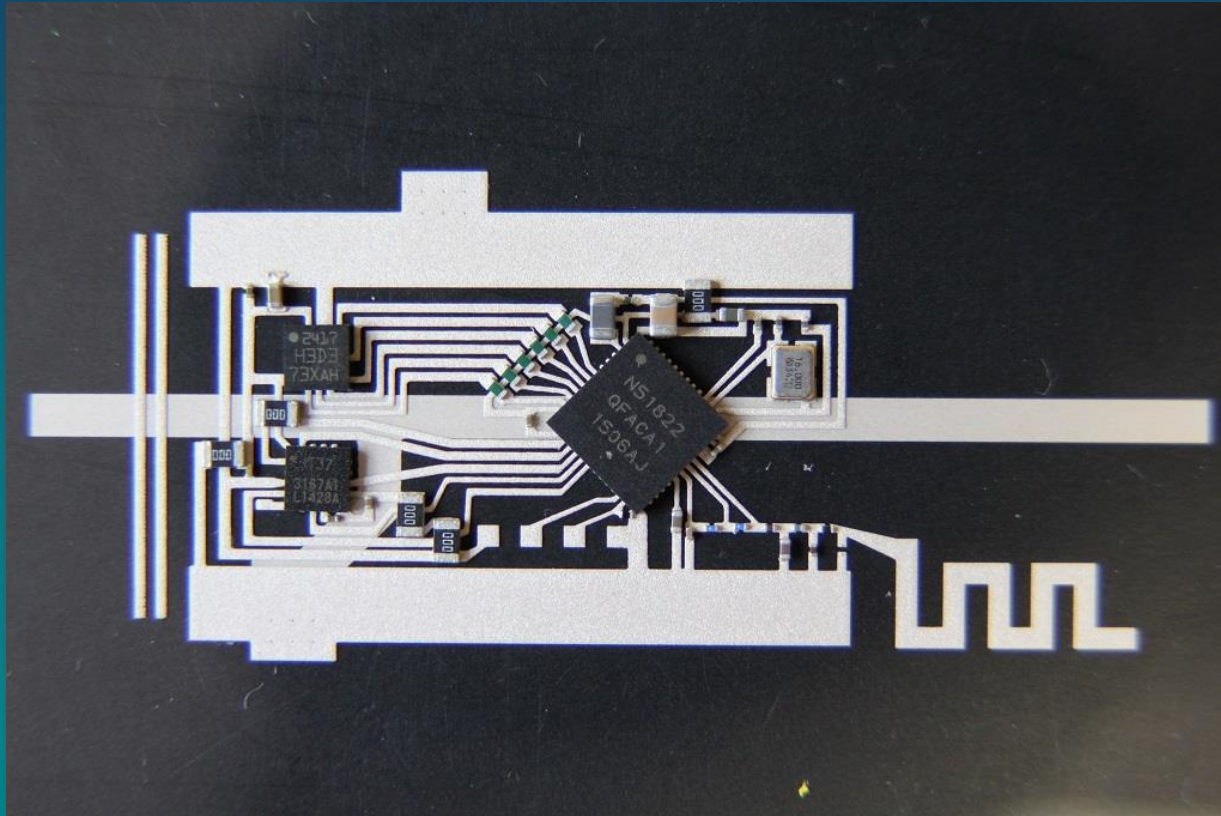
- **Challenges**

- Little industry structure
- Vast scatter of disparate technologies and technology providers
- Challenging for potential customers



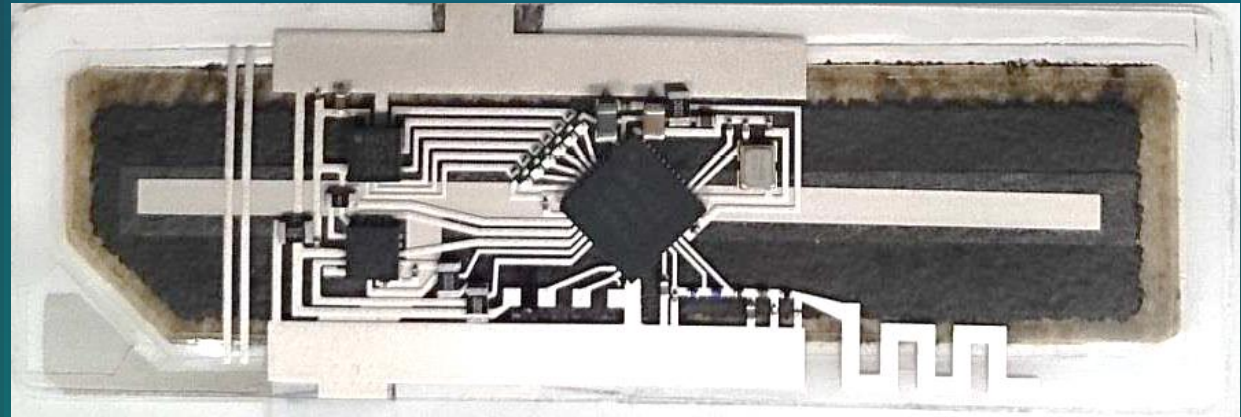
Sensor Platform

Integrated hybrid electronics product



Design objectives

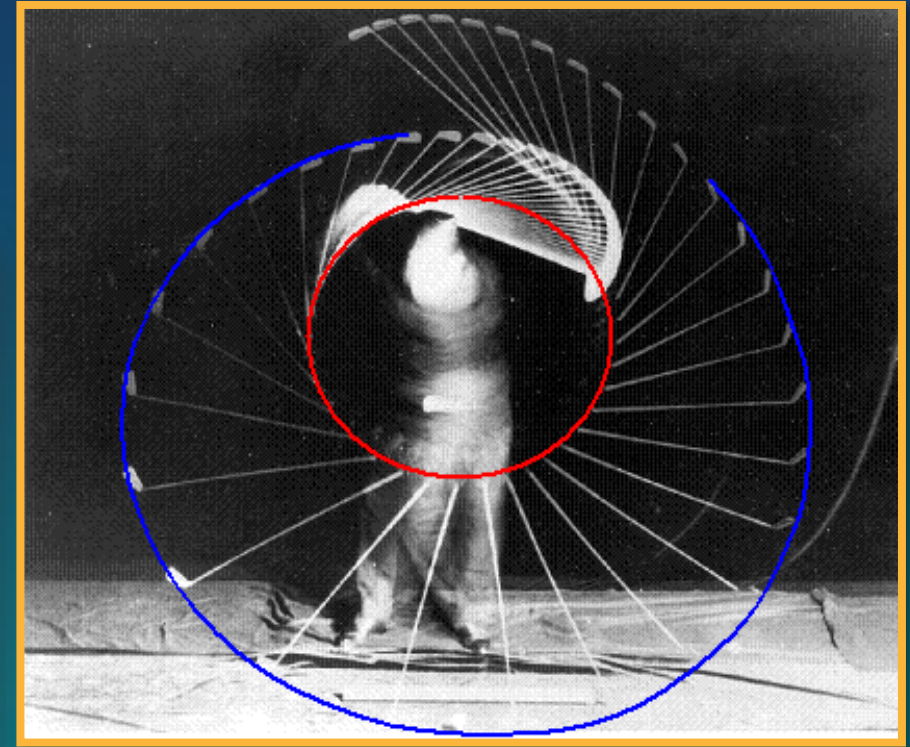
- Robust 3-axis gyroscope
- Robust 3-axis accelerator
- Disposable (1 hr to 7 days lifetime)
- Band-Aid like form factor and application



Electronics and battery form a single monolithic device

Sensor Platform

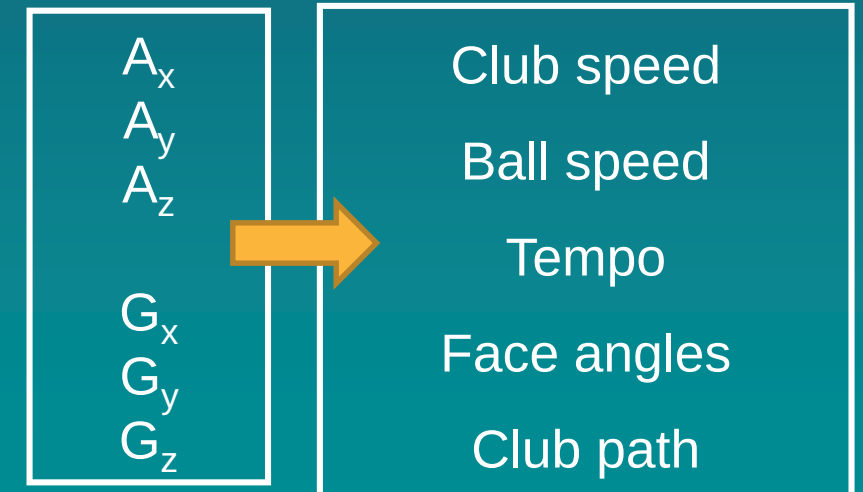
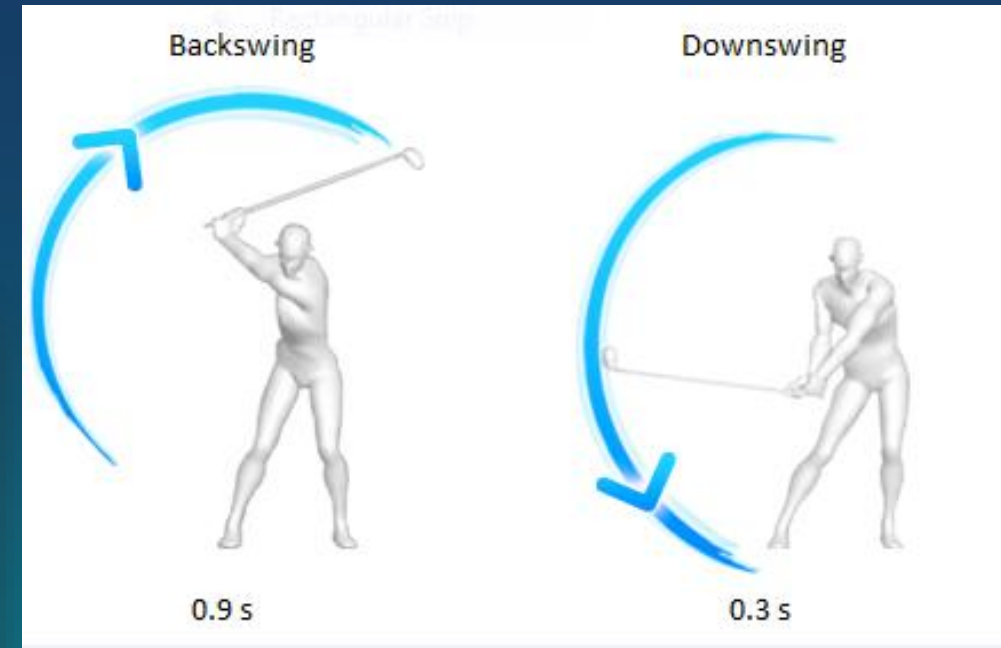
Golfing application



120 mph club head velocity
250 G centripetal acceleration
3000°/s rotation
1500 G shock load
160 mph ball launch velocity

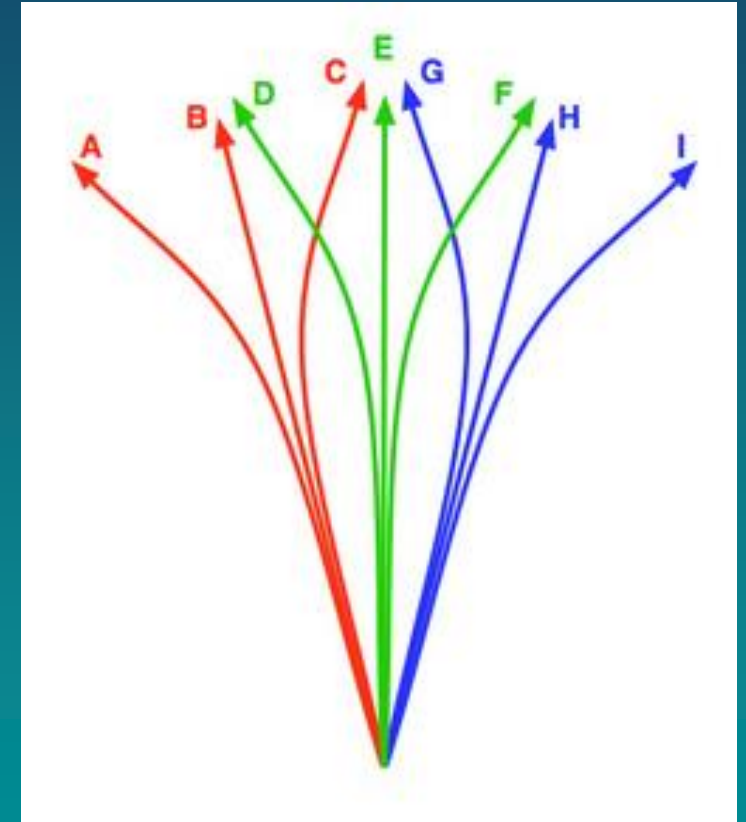
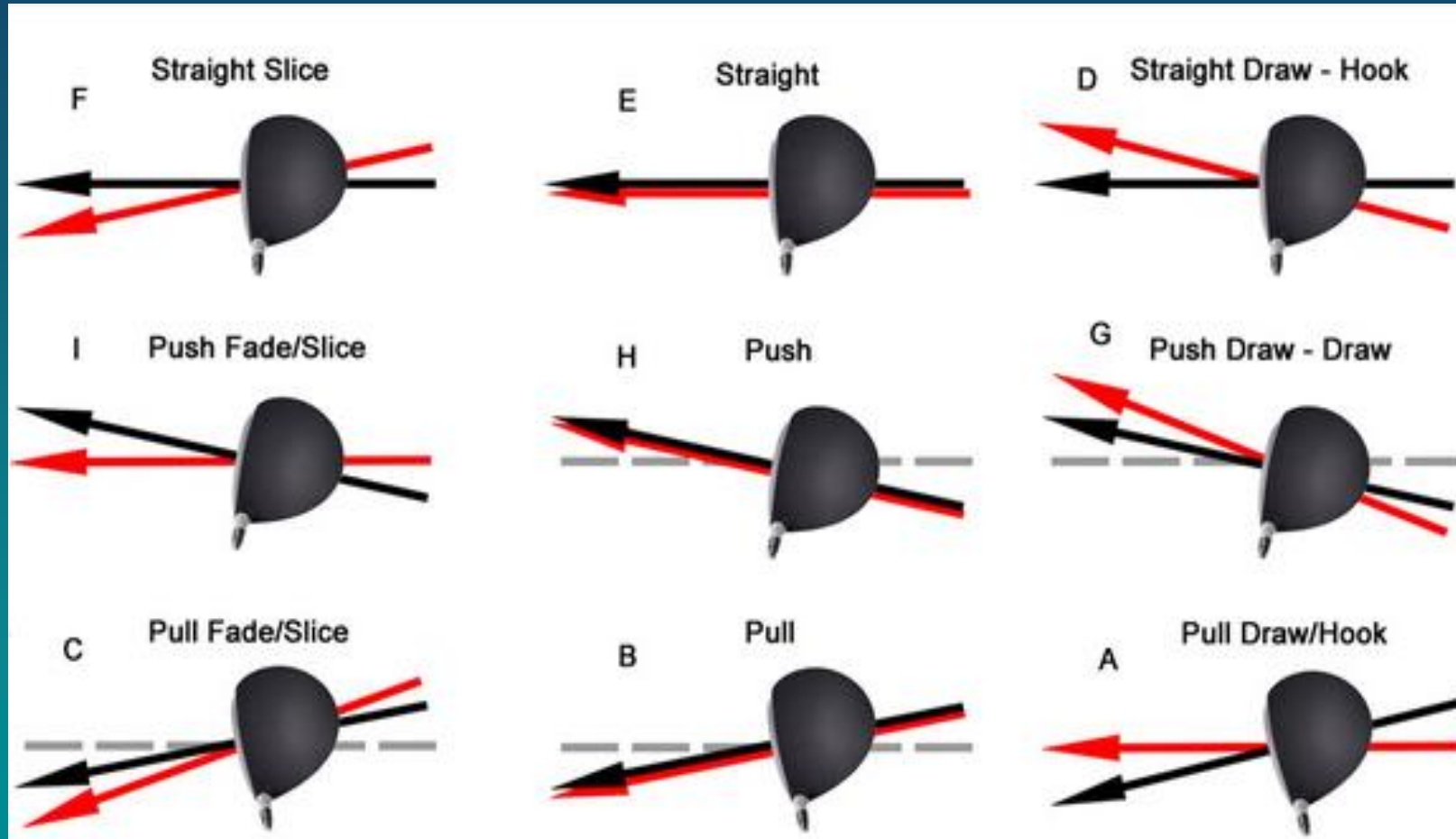
Sensor Platform

Golfing application



Golf Sensor Patch

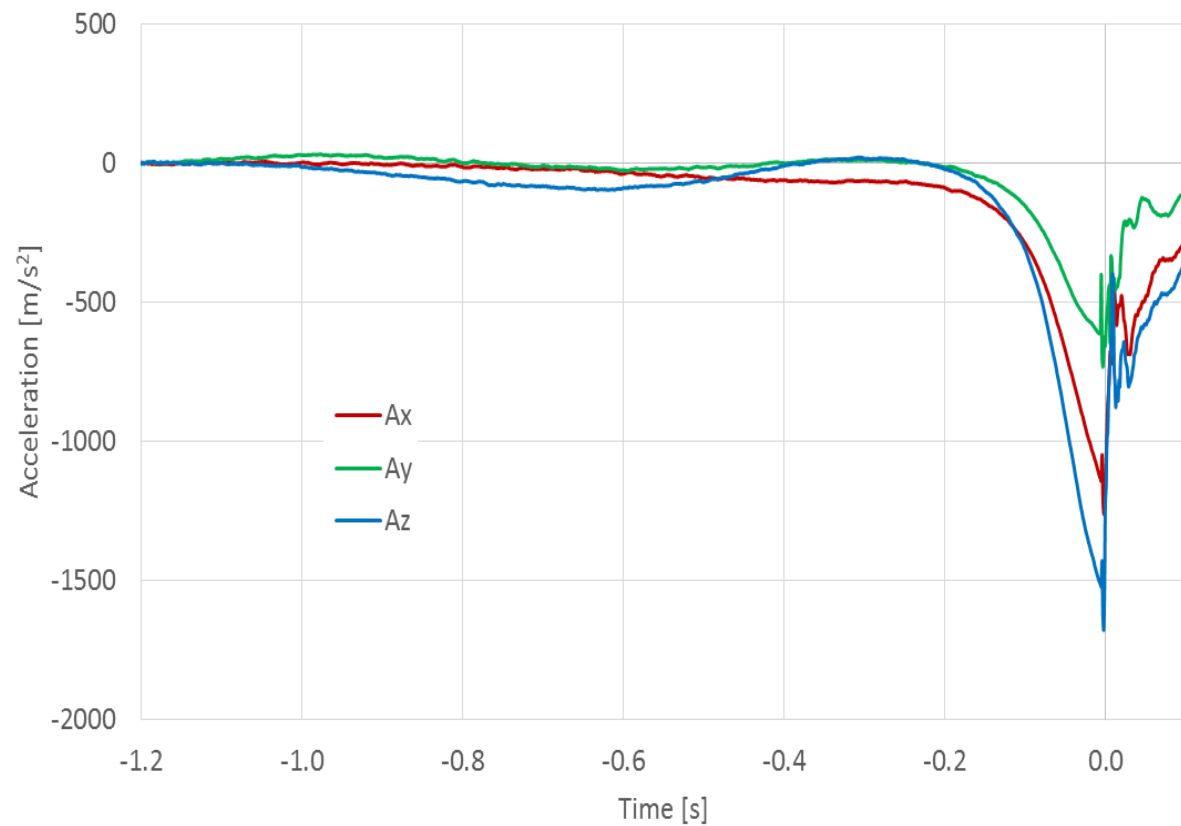
Measuring acceleration (A) and angular velocity (G) during swing



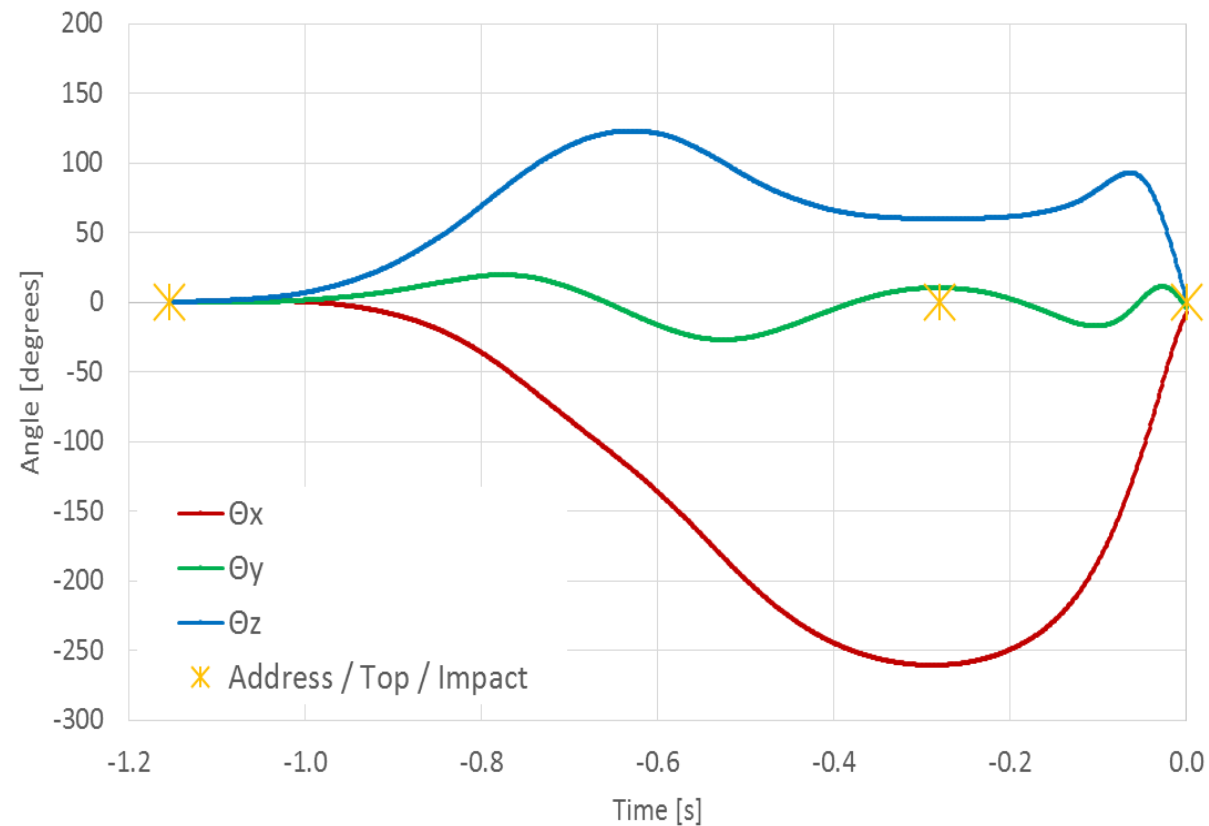
Black: Club Face

Red: Club Path

Local acceleration vs. time



Global angles



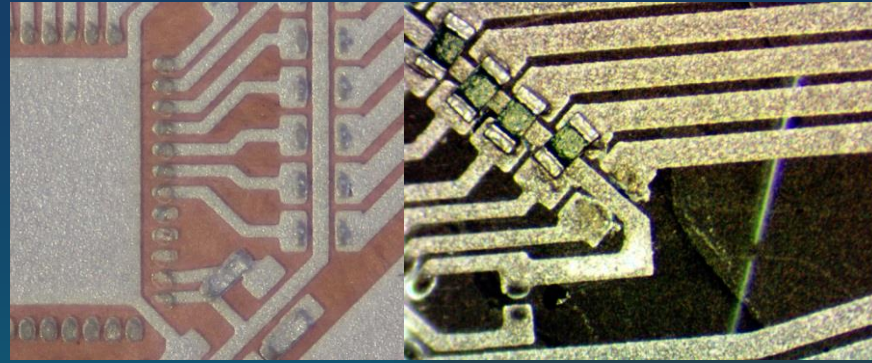
Initial Results: Velocity

Measurements show excellent precision and accuracy

Strike	Calculated Velocity [mph]	Reference Value [mph]	Error [%]
1	114.3	115.9	-1.4
2	111.4	112.4	-0.9
3	120.2	119.6	0.5
4	118.5	119.2	-0.6
5	119.0	119.0	0.0
6	125.0	122.2	2.3
7	120.8	120.4	0.4
8	122.1	122.9	-0.7
9	121.7	122.6	-0.7
10	119.2	120.4	-1.0
11	120.6	120.1	0.4
12	119.2	120.0	-0.7
13	121.5	120.1	1.2

Challenges

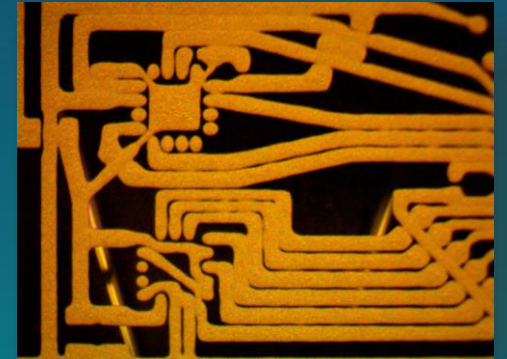
- Circuit pattern printing
- Electrical characteristics
 - Materials performance vs. processability vs. price
 - Rigid board vs. prototype product vs. actual product
- Mechanical stress
- Design for mass manufacturability
- Cost targets



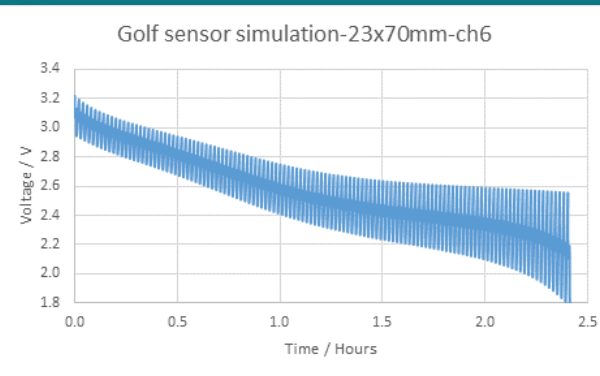
Mechanical stress



Inkjet print



Screen print



Electrical and electromagnetic characteristics



THE INTERNET OF EVERYTHING

25 BILLION

connected devices by 2020

Source: Machina Research, Feb '14

Thank you

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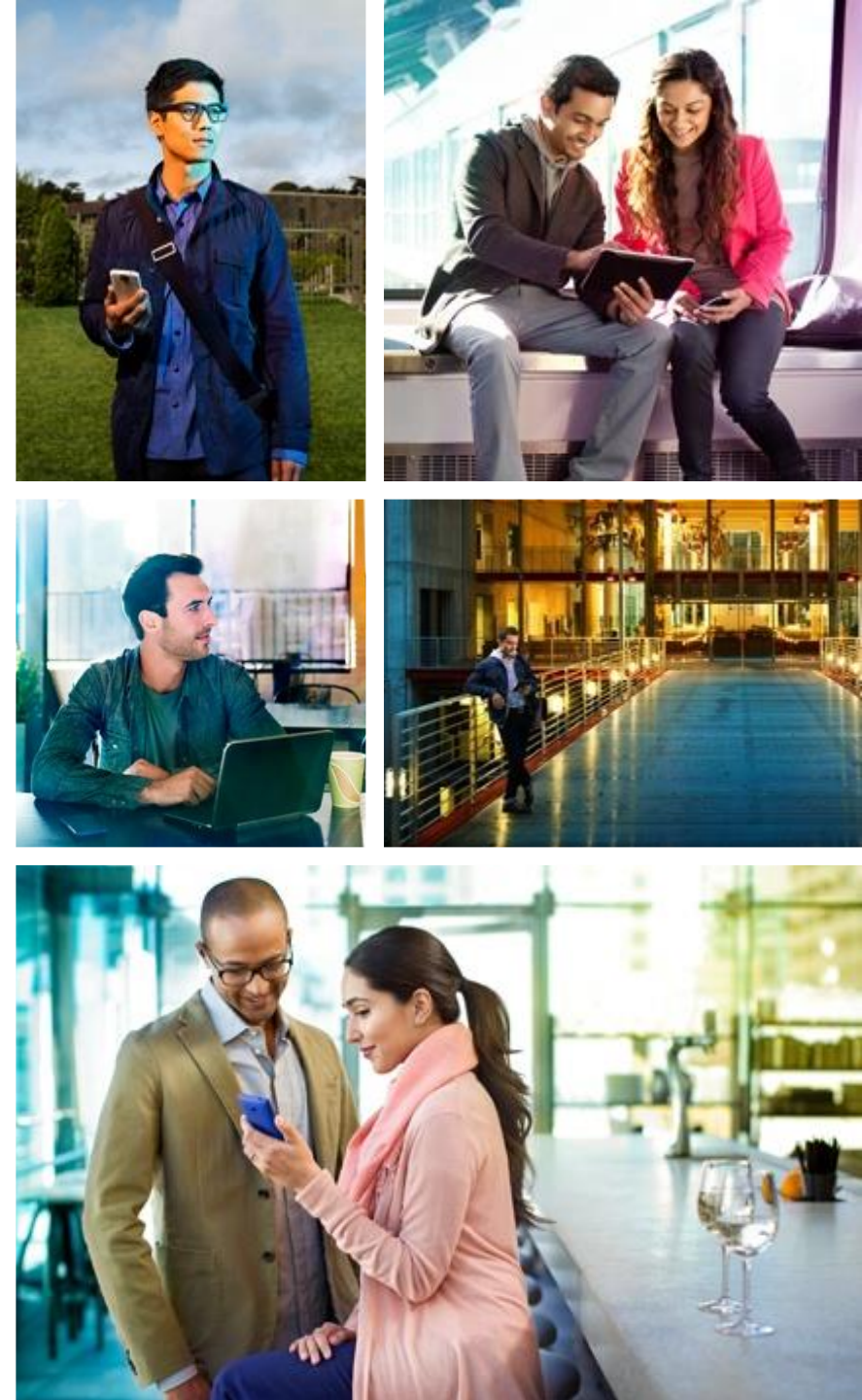
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Born Mobile™

- 29+ years of driving the evolution of wireless communications
- Making wireless more personal, affordable and accessible
- World's largest fabless semiconductor company
- S&P 100/ S&P 500/ Fortune 500



Qualcomm fuels major technology shifts in the industry

Anticipating the big challenges and investing early on to solve them

>\$34B Cumulative R&D

Digitized mobile communications

From analog to digital

- 1989: Qualcomm proves CDMA
- 1993: Demonstration of TCP/IP over CDMA
- 1998: First High Data Rate (HDR) demo
- 1999: CDMA selected as 3G standard
- 2003: First WCDMA multimode chipset supporting HSDPA

Redefined computing

From desktop to smartphones

- 1998: First commercial CDMA smartphone
- 2001: Introduced Brew™ platform
- 2005: Scorpion CPU development initiated
- 2007: First 1 GHz mobile chip sampled
- 2008: First Android smartphone
- 2009: First 3G/LTE integrated chips
- 2013: First LTE Advanced carrier aggregation smartphone
- 2014: First 64-bit 3G/LTE integrated chip

Transforming the edge of the Internet

- LTE/Wi-Fi convergence
- LTE-U
- Machine learning
- Computer vision
- Security and privacy
- 5G

Drivers for Change

Cost is a high value driver for surface electronics (i.e., printed, deposited, flexible, etc.)

